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Application of Biomarkers in the Analysis of Sublethal Exposures to Different Metals in an Amazonian Ostracod, *Strandesia Rondoniensis*

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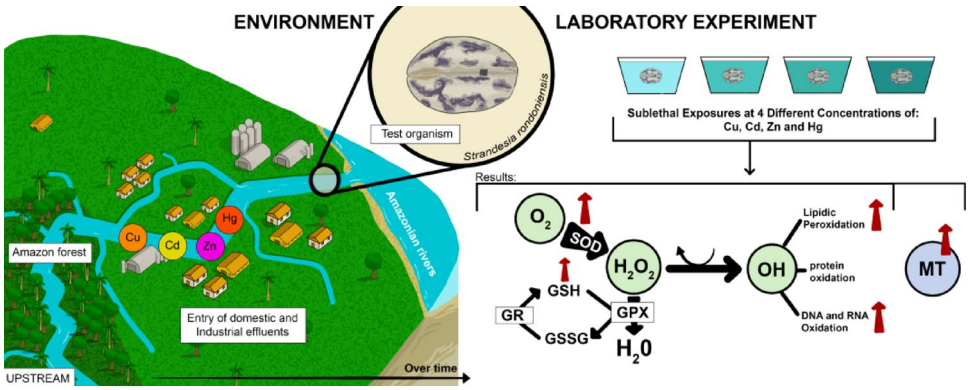
Abstract

Metals reach aquatic ecosystems through various anthropogenic activities, posing risks to aquatic life. This study investigated oxidative stress and detoxification biomarkers in Strandesia rondoniensis, an Amazonian ostracod, after 96 h of exposure to sublethal concentrations of $CuSO_4$, $ZnCl_2$, $CdCl_2$, and $HgCl_2$. Significant increases in superoxide dismutase, glutathione, and metallothionein activity were observed in response to Zn, Cd, and Hg, indicating the induction of reactive oxygen species (ROS). In contrast, copper exposure did not alter biomarker activity. Lipid peroxidation increased following exposure to Zn, Cd, and Hg, damaging cell membranes and leading to DNA damage, suggesting the antioxidant defense system was insufficient to counteract oxidative stress. The results demonstrate the sensitivity of *S. rondoniensis* to environmentally relevant metal concentrations, supporting its application as a neotropical model organism in ecotoxicological assessments. Furthermore, the findings underscore the need to revise Brazil's environmental legislation on permissible metal concentrations in aquatic environments.

Graphic Abstract



Keywords Amazon · Metals · Chronic · Oxidative damage · Native species

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Introduction

The Amazon is one of the world's most biodiverse biomes (Rico et al. 2022). However, in recent decades, icreasing population density in cities of all sizes has led to the steady discharge of domestic and industrial effluents into the Amazon's aquatic ecosystems (Gomes et al. 2023), mainly due to limited access to basic sanitation in Northern Brazil (Brazil, 2023). Moreover, illegal mining activities have increased metal inputs into the region's aquatic ecosystems (Moulatlet et al. 2023).

Metals that enter the aquatic ecosystem can cause serious damage to biota (Kahlon et al. 2018; Jeong et al. 2023). Even at sublethal concentrations, they can interact with aquatic organismsand bioaccumulate (Ali et al. 2019; Kuehr et al. 2021), leading to adverse effects on growth and reproduction (Liang et al. 2022; Rebolledo et al. 2022), immunotoxicity (Ray et al. 2015), and oxidative stress due to increased production of reactive oxygen species (ROS) (Watson et al. 2018; Jeong et al. 2023). ROS can damage various macromolecules, including, lipids, proteins, and DNA (Jeong et al. al. 2023).

However, cells have an efficient antioxidant defense system, both enzymatic (e.g. superoxide dismutase – SOD; catalase – CAT; glutathione peroxidase – GPx; among others) and non-enzymatic (e.g. glutathione – GSH) to cope with those metabolites. Under normal physiological conditions, the adverse effects of ROS are neutralized by the defense system (Zheng et al. 2022), which inhibit or reduce their cellular damage. However, if the system fails, lipid peroxidation (LPO), one of the main mechanisms of oxidative injury, is used as a biomarker of effects in different organisms (Timme-Laragy et al. 2024).

From this perspective, the use of different biomarkers in environmental monitoring represents progress toward understanding the effects of metal pollution (Kumar et al. 2024). Although biomarkers are well established as effective tools for verifying adverse effects across different trophic levels (Lee et al. 2021; Kim et al. 2022), studies applyig them to aquatic invertebrates - especially ostracods - remain scarce. The literature reports only one study identifying genes linked to antioxidant in the ostracod *Heterocypris incongruens* (Hiki et al. 2019). Therefore, assessing the antioxidant defense and oxidative damage systems in invertebrates is essential to develop appropriate methods for early detection of chemical effects on aquatic biota (Kumar et al. 2024).

Thus, this study assessed enzymatic and non-enzymatic biomarkers related to detoxification, oxidative defense, oxidative stress, and DNA damage, including SOD, GSH, metallothionein (MT), LPO, and DNA strand break markers, in the species *S. rondoniensis* recently described by Gomes et al. (2024a), following sublethal exposure to environmentally relevant concentrations of metallic salts copper sulfate, zinc, cadmium, and mercury chloride.

Materials and Methods

Test Organism Sampling and Maintenance

Organisms were collected in Porto Velho Natural Park, Rondônia State (8º 41' 13.26" S; 63º 52' 10.06" W), in August 2018, under SISBio authorization (Nº:67974-1), and transferred to the Laboratory of Limnology and Aquatic Ecotoxicology at the Federal University of São Carlos (UFSCar) for cultureestablishment, following adaptations of the protocol for tropical cladoceran species *Ceriodaphnia sp.* (ABNT, 2022, Nº 13373). Reconstituted water was prepared with the following parameters: pH of 7-7.6, hardness as $CaCO_3$ at 40–48 mg L^{-1}, and conductivity of 160 µS cm^{-1} maintained under controlled temperature (25 °C ± 1 °C) and light regime (12 h light / 12 h dark). The diet consisted of 20 mL Tetramin® fish food (5 g L^{-1}) and 20 mL *Raphidocelis subcapitata* suspension (≈ 10^8 cells L^{-1}). Detailed taxonomic and life cycle descriptions were provided by Gomes et al. (2024a) and the predicted sensitivity of the species was reported by Gomes et al. (2024b).

Preparation and Quantification of Stock Solutions

The following metal salts were used to evaluate the chronic effects of metals: copper sulfate ($CuSO_4$), cadmium chloride ($CdCl_2$), zinc chloride ($ZnCl_2$), and mercury chloride ($HgCl_2$). Stock solutions of each metal salt were prepared in distilled water at a concentration 10 mg L^{-1}. Nominal test concentrations were obtained by diluting the stock solutions in reconstituted water. Aliquots of the stock solutions were acidified with analytical-grade HNO_3 to a pH < 2.0 and used to determine actual concentrations of $CuSO_4$, $CdCl_2$, and $ZnCl_2$. Analyses were conducted using an Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES Optima 8300, Perkin Elmer) following the 23rd edition of Standard Methods (SMWW, 3111 B). $HgCl_2$ was quantified using a DMA-80 EVO Milestone mercury analyzer (USEPA, 7473).

Exposure of Test Organisms to Metals

Three concentrations of each metallic salt were prepared and set as the no-observable effect concentration (NOEC) from the acute toxicity values presented by Gomes et al. (2024b). Only environmentally relevant concentrations were tested: $CuSO_4$ (0; 0.001; 0.01; 0.1 mg L^{-1}), $ZnCl_2$ (0;

0.01; 0.1; 1 mg L^{-1}), $CdCl_2$ (0; 0.001; 0.01 mg L^{-1}), and $HgCl_2$ (0; 0.001; 0.01; 0.05 mg L^{-1}). Exposures were conducted in 250 mL non-toxic plastic containers with 100 mL of the test solution and only one experiment was carried out for each metal, with five replicates per concentration with 150 adult individuals each. The experiment lasted 96 h and was conducted at a controlled temperature of 25 ± 1 °C in the dark. Food (2 mL of a fish food solution) was provided at the beginning and after 48 h. Water parameters, including, pH, electrical conductivity (EC), temperature (T), and dissolved oxygen (DO) were measured at the beginning and end of the experiment.

Biomarker Analysis

After 96 h-exposure, the biomass (0.001 mg) of the pool of all individuals from each replicate ($n = 150$) was determined, and samples were immediately stored at −80 °C for molecular and biochemical analyses. Samples were then homogenized by ultrasound (Eco-Sonics, QR500, Brasil) in a 0.1 M phosphate buffer (7.6 pH) and centrifuged at 10.000 × g for 10 min at 4 °C. A 1:5 buffer to sample mass ratio was used.

Protein concentration (pt, mg mL^{-1}) was quantified using the Bradford method (1976), with bovine serum albumin as standard; and readings were taken at 595 nm. Table S1 shows the protein concentration for each sample used to standardize the enzyme activity.

The metallothionein (MT) concentration was determined by Tris-HCl 20 mM buffer with 0.5 mM sucrose and B-mercaptoethanol (Viarengo et al. 1997). The quantification of the enzymatic activity of superoxide dismutase (SOD, U mg pt^{-1} min^{-1}) was based on the inhibition of cytochrome C reduction rate by superoxide radicals at 550 nm, according to the method described by McCord and Fridovich (1969). Glutathione (GSH, nmol GSH mg pt^{-1} min^{-1}) concentration was quantified by the reaction of 2,3- naphthalenedicarboxaldehyde (NDA) with GSH, forming a fluorescent compound measured at excitation/emission wavelengths of 472/528 nm (White et al. 2003).

Lipid peroxidation (LPO, nmol CHP mg pt^{-1}) was quantified using the xylenol orange ferrous oxidation method (FOX), mesuring absorbance at 560 nm, following Jiang et al. (1991, 1992). DNA strand breaks (µg DNA mg pt^{-1}), which indicate oxidative stress-related DNA damage, were quantified following the alkaline precipitation protocol (Olive 1988). All analyses were performed using a SpectraMax M5 microplate reader (Multi-Mode, Molecular Devices, USA).

Statistical Analysis

Normality, homogeneity and significant differences in each biomarker between treatments were assessed by comparing the control and evluated by Generalized Linear Models (GLM) with a Gaussian family and identity link-function in Jamovi software, version 2.4 (Gallucci 2019). Statistical significance was determined at a 95% confidence level with p-values < 0.05.

Results

Survival

Exposure to any tested concentrations of $CuSO_4$, $ZnCl_2$, and $HgCl_2$ had no effect on *S. rondoniensis* survival after 96 h ($p > 0.05$). On the other hand, exposure to $CdCl_2$ at 0.1 mg L^{-1} reduced survival compared with the control. Consequently, biomarkers were not measured in that treatment, as the objective was to assess physiological responses.

Water Parameters of Test Solutions

The water parameters at beginning and end of the experiments remained stable, with pH ranging from 7.4 to 7.8, electrical conductivity from 133.4 to 216 µS cm^{-1}, dissolved oxygen from 4.3 to 6.9 mg L^{-1}, temperature from 24.5 to 25.3 °C, and hardness from 40 to 48 mg $CaCO_3$ L^{-1} (Table S2). The concentrations of metals adopted did not differ by more than 10% from the nominal values (Table S3); therefore, nominal concentrations are reported throughout the results, as recommended by the ISO standard (2000).

Detoxification Biomarker and Antioxidant Defense Biomarkers

Regarding the control, MT levels were significantly higher in individuals exposed to Zn, Cd, and Hg at concentrations of 0.01 and 1 mg L^{-1}, 0.001 mg L^{-1}, and 0.05 mg L^{-1}, respectively ($p < 0.05$, Fig. 1), data are presented in supplementary material Table S4.

SOD activity was significantly elevated relative to the control group ($p < 0.05$) at 0.01 mg L^{-1} of $CdCl_2$ and 0.05 mg L^{-1} $HgCl_2$, whereas no significant effects were observed for Cu and Zn salts (Fig. 2, $p > 0.05$). Conversely, GSH content increased in *S. rondoniensis* individuals exposed to $ZnCl_2$, $CdCl_2$, and $HgCl_2$ ($p < 0.05$, Fig. 3) at 1, 0.01, and 0.01 concentrations, respectively. For $HgCl_2$, only two treatments were analyzed as the available sample amount for the 0.01 and 0.05 mg L^{-1} groups was insufficient.

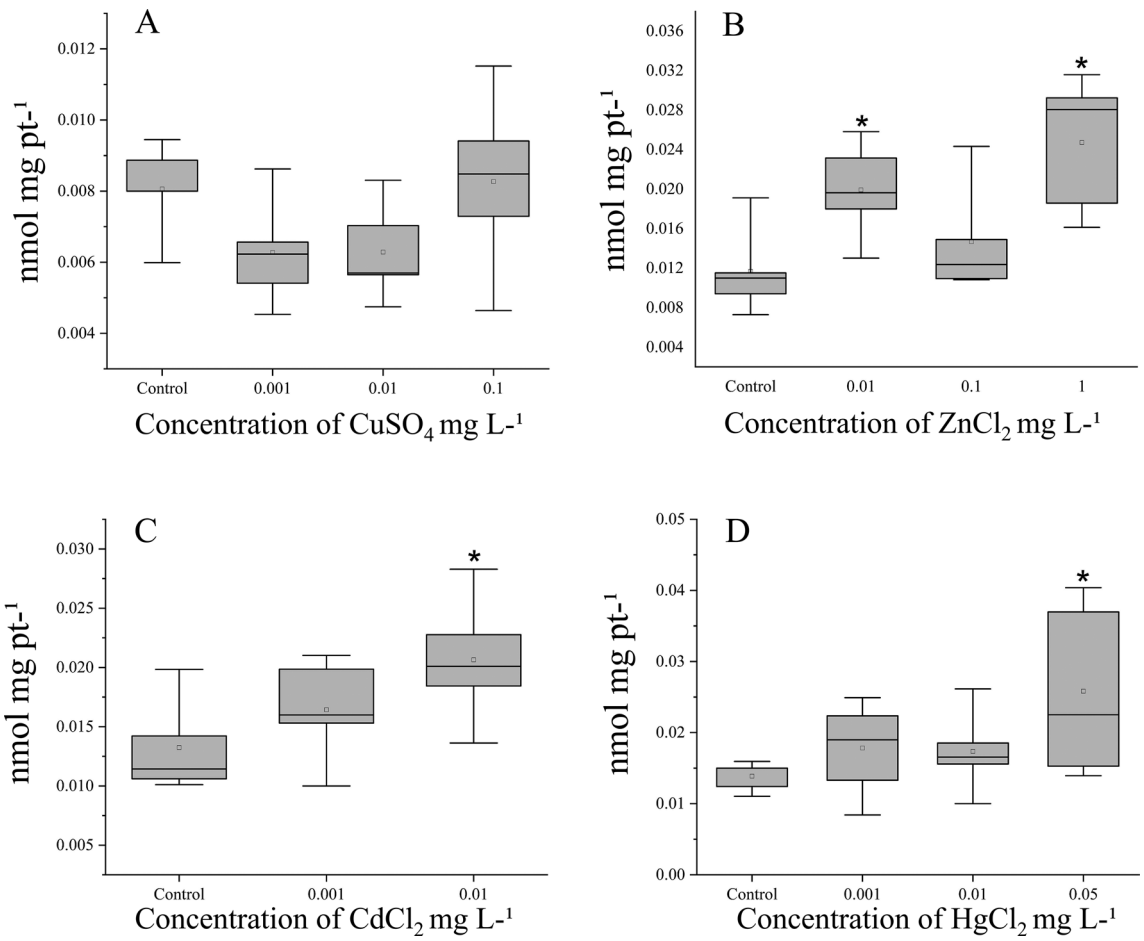

Fig. 1 Metallothionein (MT) (Data in Mean - Minimums and Maximums) quantification in *Strandesia rondoniensis* exposed to different metals (**A** – $CuSO_4$; **B** – $ZnCl_2$; **C** – $CdCl_2$; **D** – $HgCl_2$) for 96 h. Asterisks (*) indicate statistically significant differences from the control ($p<0.05$)

Stress and Oxidative Damage Biomarkers

Lipid peroxidation (LPO) levels showed a significant increase ($p<0.05$) after exposure to $ZnCl_2$ (1 mg L^{-1}), $CdCl_2$ (0.01 mg L^{-1}), and $HgCl_2$ (0.01 and 0.05 mg L^{-1}) (Fig. 4).

DNA strand break analysis revealed that exposure to Zn, Cd, and Hg caused significant increases ($p<0.05$) in breaks relative to the control at concentrations of 1 mg L^{-1} $ZnCl_2$, 0.001 and 0.01 mg L^{-1} $CdCl_2$, and 0.05 mg L^{-1} $HgCl_2$ (Fig. 5).

Discussion

Detoxification and Antioxidant Defense Mechanisms

Metal pollution in water and sediment has raised concerns in the scientific community for decades. It is well established that metals increase the production of reactive oxygen species (ROS), such as hydroxyl radicals (•OH) and hydrogen peroxide (H_2O_2), due to the uncoupling of the Haber-Weiss and Fenton reactions. This uncoupling results from the interaction ionic metals with molecular oxygen (Soriano-Agueda et al. 2016) leading to oxidative stress (Ren et al. 2019).

Exposed organisms develop antioxidant defense mechanisms to protect cells against ROS attack. These mechanisms include enzymes such as superoxide dismutase (SOD) which converts superoxide radicals (O_2^-) into hydrogen peroxide (Jiang et al. 2024). This peroxide is then converted into water and oxygen by other enzymes in the antioxidant defense cascade (Jiang et al. 2024). SOD activity represents one of the main antioxidant enzyme systems in aquatic invertebrates (Zheng et al. 2022) and serves as the first line of defense against xenobiotic toxicity (Kim et al. 2015). In addition, GSH is an important substrate for antioxidant and detoxification enzymes as glutathione S-transferase (GST), glutathione peroxidase (GPx), and glutathione reductase (GR) (Scopetani et al. 2020).

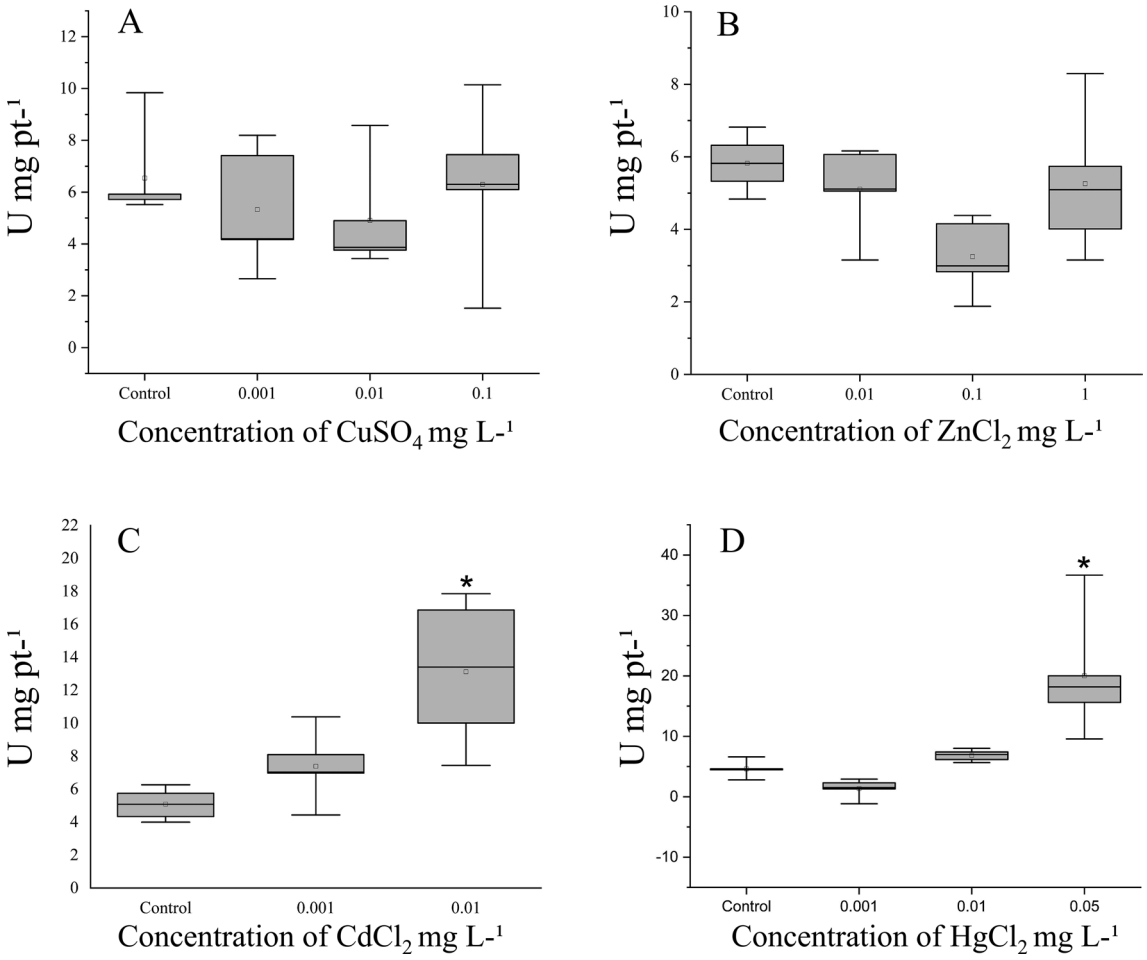

Fig. 2 Superoxide dismutase (SOD) (Data in Mean - Minimums and Maximums) enzymatic activity (U mg pt^{-1}) in *Strandesia rondoniensis* exposed to different metals (**A** – $CuSO_4$; **B** – $ZnCl_2$; **C** – $CdCl_2$; **D** – $HgCl_2$) for 96 h. Asterisks (*) indicate statistically significant differences from the control ($p < 0.05$)

According to our findings, Cd and Hg increased SOD levels, while Zn, Cd, and Hg increased GSH in exposed individuals. The rise in SOD is linked to elevated ROS caused by metal toxicity (Zheng et al. 2022; Ray et al. 2015). A study conducted by Dong et al. (2024) demonstrated that GSH levels in gill cells can increase after prolonged exposure to metals such as cadmiun, and the authors stated that this rise in GSH production act as a compensatory response to metal-induced oxidative stress, supporting antioxidant enzyme and enhancing cellular detoxification capacity, since GSH plays a fundamental role in non-enzymatic antioxidant defense (Mahboob 2013).

In line with this mechanism, Ren et al. (2019) reported that Cd concentrations of 5 to 50 µg L^{-1} increased both GSH content and SOD activity in the crustacean *Marsupenaeus japonicus*, while Chen et al. (2014) reported Hg led to an increased expression of genes linked to SOD in sublethal exposures to *Venerupis philippinarum* mollusk. Therefore, data from this study reveal that the increase in SOD levels, associated with the rise in GSH, mainly due to exposure to Zn, Cd, and Hg, shows that *S. rondoniensis* experienced elevated ROS production, stimulating antioxidant defense systems.

Metallothioneins (MT) are non-enzymatic proteins that bind metals via thiol groups on cysteine residues, playing a key role in protecting organisms from metal toxicity, while contributing to the metabolism and homeostasis of essential metals (Oros et al. 2025). They have been found in several invertebrates, including mollusks and crustaceans (Gora et al. 2025). In this study, Zn, Cd, and Hg were found to increased MT concentrations, as already reported for other organisms exposed to metals. According to Gunderson et al. (2021), low concentrations of Zn increased MT concentrations in *Pacifastacus leniusculus* crayfish and Verlecar et al. (2008) observed a significant increase in the MT level in the digestive gland of *Perna viridis* mussel exposed to 45 µg L^{-1} of Hg. Furthermore, exposure to 20 µg L^{-1} of Hg induced a significant increase in MT in the gills of *Crassostrea gigas* oyster and *Mytilus edulis* mussels in 21 days of exposure (Géret et al. 2002). On the other hand, Cd also induced MT

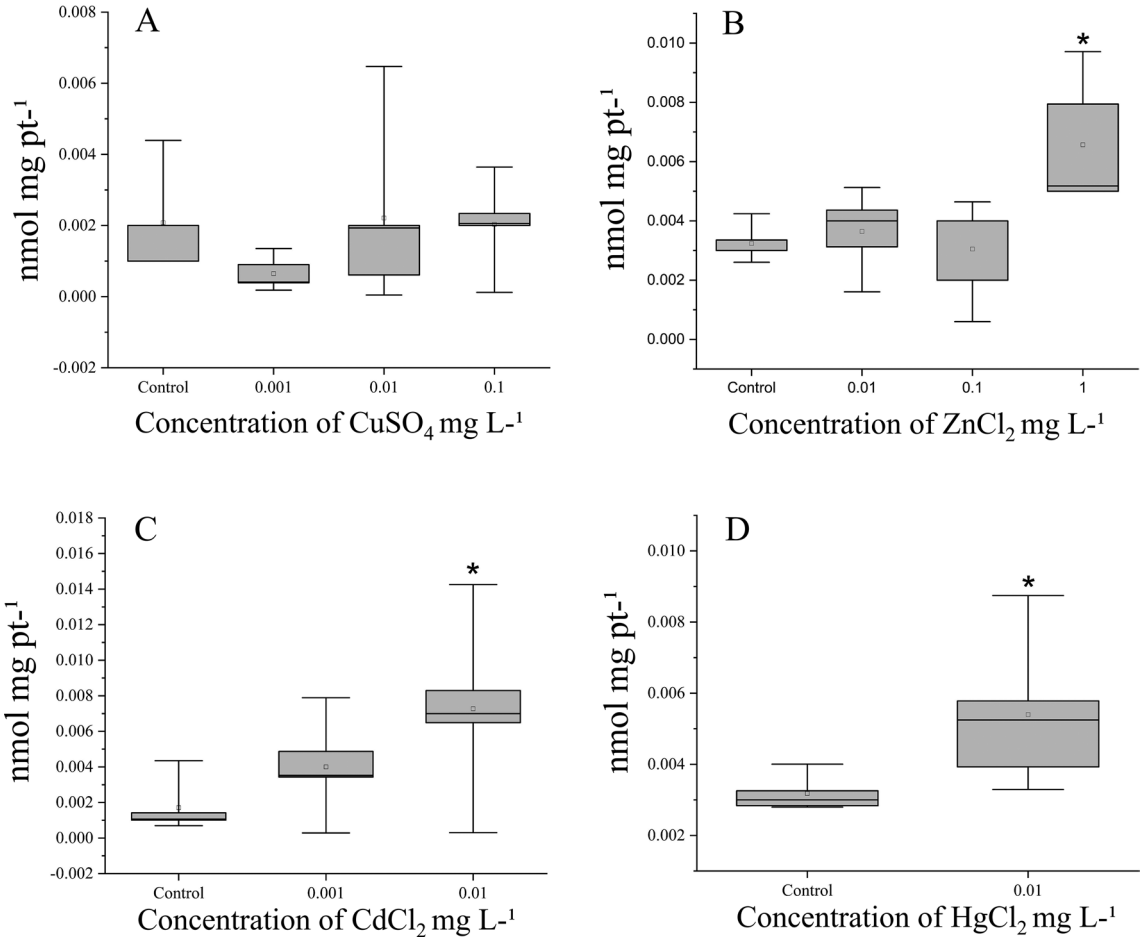

Fig. 3 Non-enzymatic antioxidant glutathione (GSH) (Data in Mean - Minimums and Maximums) concentration (nmol mg pt^{-1}) in *Strandesia rondoniensis* exposed to different metals (**A** – $CuSO_4$; **B** – $ZnCl_2$; **C** – $CdCl_2$; **D** – $HgCl_2$) for 96 h. Asterisks (*) indicate statistically significant differences from the control ($p<0.05$)

expression in the gills of *Crassostrea virginica* oyster after 4 days of exposure to 50 µg Cd L^{-1} (Roesijadi e Klerks, 1989). Therefore, the data provided in this study revealed *S. rondoniensis* exposed to different metals also increased its MT levels in response to the concentrations tested through that mechanism as an attempt to detoxify the metals.

Stress and Oxidative Damages

Lipid peroxidation (LPO) occurs when excessive ROS accumulated in cells, interacting with membrane lipids and triggering the production of lipid peroxides, that disrupt normal cell function (Regoli and Giuliani 2014). It is one of the main biomarkers of oxidative stress, indicating the loss of key cellular functions, such as disturbances in membrane permeability, ionic flow, impaired nutrient and toxic exchange, lipoprotein oxidation, and DNA damage (Hariharan et al. 2012; Mohamood et al. 2021).

In general, organisms from polluted environments show increased LPO values (Righi et al. 2022). In the present study, this increase indicates oxidative stress induced by Zn, Cd, and Hg in *S. rondoniensis*. On the other hand, Cu did not cause significant changes in any of the biomarkers analyzed. Lipid oxidation damage various physiological processes, as addressed elsewhere. The LPO response observed here aligns with findings by Oliveira et al. (2018), who reported increase LPO in the bivalve *Corbicula fluminea* exposed to mercury. Elevated LPO was observed in *Callinectes ornatus* and *C. bocourti* from more impacted areas with higher Cd concentrations, indicating oxidative stress linked to environmental contamination (Righi et al. 2022), and Zn caused an increase in LPO after 30 days of exposure at 0.18 mg L^{-1} in the crustacean *Penaeus monodon* (Hariharan et al. 2012).

DNA damage is widely used as a biomarker to assess metal genotoxicity (Bolognesi e Cirillo, 2014), as by significant DNA strand breaks in *S. rondoniensis* exposed to elevated concentrations of Zn, Cd, and Hg.This DNA damage response aligns with the other findings in the present study, where increased levels of SOD, GSH, and MT indicate that the organisms activated mechanisms to mitigate

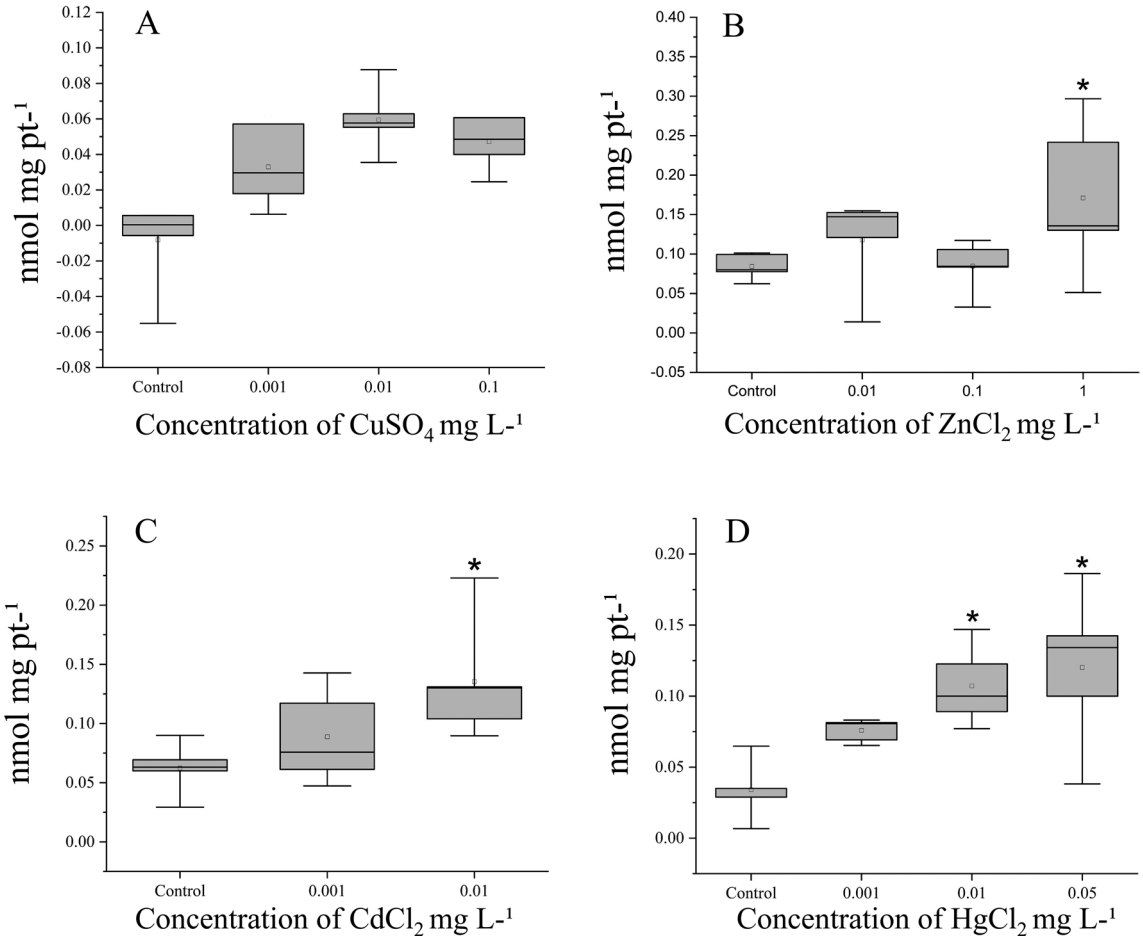

Fig. 4 Lipid peroxidation (LPO) (Data in Mean - Minimums and Maximums) responses in *Strandesia rondoniensis* exposed to different metals (**A** – $CuSO_4$; **B** – $ZnCl_2$; **C** – $CdCl_2$; **D** – $HgCl_2$). Asterisks (*) indicate statistically significant differences from the control ($p<0.05$)

ROS generated during metal detoxification. However, despite these defense responses, the antioxidant system was insufficient to fully prevent oxidative damage, as evidenced by the significant increase in LPO levels, ultimately leading to DNA damage. Evidence of that process provoked by Cd had already been provided by Chandra and Bukhsh (2004), who found genotoxicity in tissues of *Oreochromis mossambicus.* Furthermore, a study with *Gammarus elvirae* amphipod showed significant DNA damage in hemocytes after 24 h-exposure to 0.1 µg L^{-1} of Hg (Di Donato et al. 2016).

The responses verified in this study are concerning, as the effects on different biomarkers occurred at concentrations within the limits established by Brazilian law, which allows effluent discharge concentrations of 0.2 mg L^{-1} for Cd, 1.0 mg L^{-1} for Cu, 5.0 mg L^{-1} for Zn, and 0.01 mg L^{-1} for Hg (Brasil, 2011). However, the legislation that set classification and environmental guidelines for water bodies is more restrictive for several metals (Brasil 2005). Gomes et al. (2023) reported that approximately 44% of all metals evaluated in Amazon waters exceeded these limits, while Moulatlet et al. (2023) found maximum metal concentrations in Amazon water samples influenced by mining activities reaching 1 mg L^{-1} for Cu, 2 mg L^{-1} for Zn, 0.043 mg L^{-1} for Cd, and 0.1 mg L^{-1} for Hg.

The aforementioned elevated metal concentrations pose risks not only to aquatic biota (Viana et al. 2021; Gomes et al. 2023), but also to human populations inhabiting the Amazon biome (Gomes et al. 2020; Galarza et al. 2023). The present results show the limits established by law must be reformulated based on concrete scientific evidence of metal presence in Amazonian aquatic ecosystems (Gomes et al. 2023; Moulatlet et al. 2023). However, biomarker studies on native to the Amazonian ostracods are lacking. This study is the first to propose *S. rondoniensis* as a potential model organism for environmental monitoring, as it exhibited clear and consistent responses across multiple biomarkers of stress and oxidative damage.

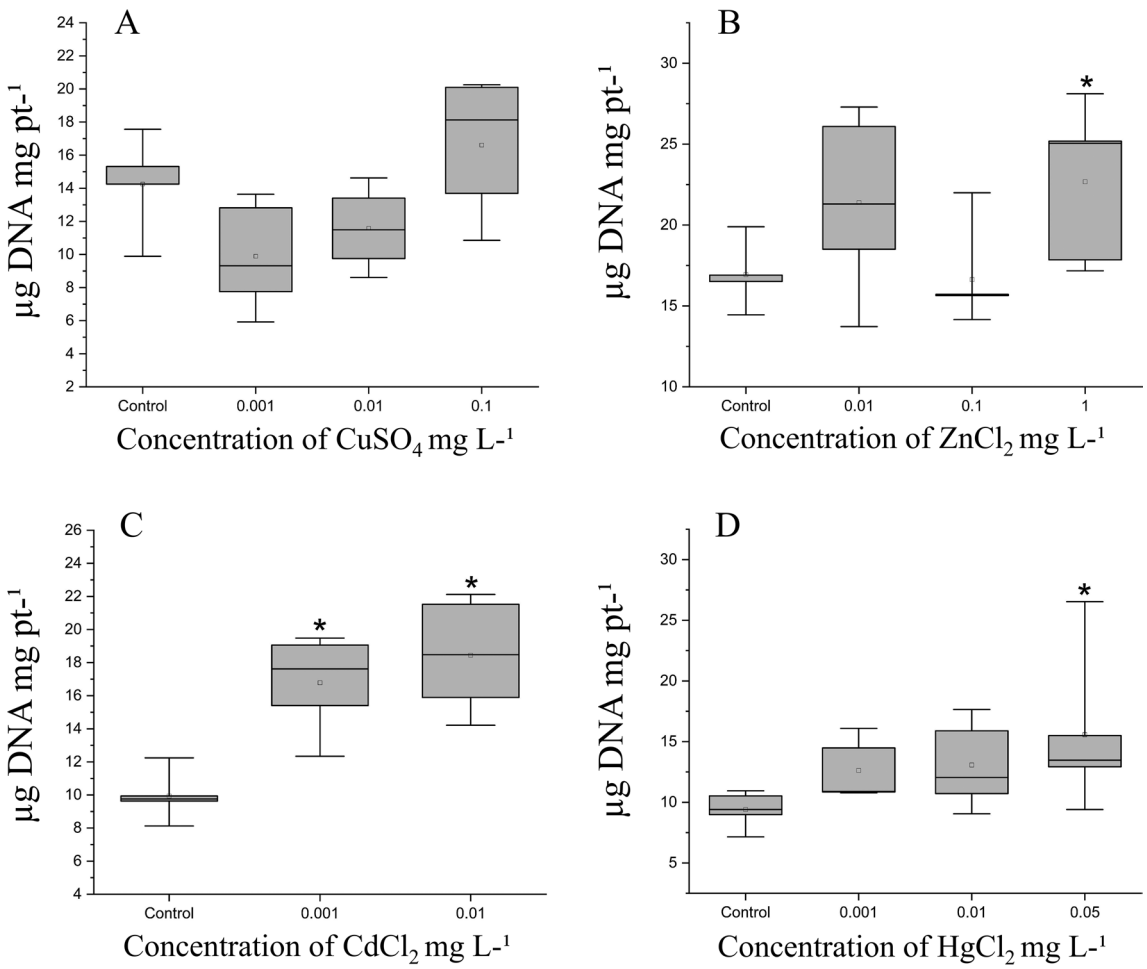

Fig. 5 DNA damage responses (Data in Mean - Minimums and Maximums) in *Strandesia rondoniensis* exposed to different metals (**A** – $CuSO_4$; **B** – $ZnCl_2$; **C** – $CdCl_2$; **D** – $HgCl_2$). Asterisks (*) indicate statistically significant differences from the control ($p < 0.05$)

Conclusion

The results of this study highlight, for the first time, the response of an Amazonian native ostracod to different biomarkers of antioxidant defense, oxidative stress, and cellular damage. I – Cu was the only metal that did not cause changes in the antioxidant defense biomarkers of the test organism. Contrarily, Zn, Cd, and Hg metal salts caused an increase in ROS in cells even at environmentally relevant low concentrations; II – The increase in LPO levels indicates that antioxidant defense systems failed to prevent damage to plasma membranes, and III –Together, these factors caused DNA damage, whichmay lead to adverse effects at both individual and population levels over time, especially such damage persistents and DNA repair mechanisms are inefficient. These findings underscore the value of biomarker application in environmental monitoring, demonstrating effects even at environmentally relevant concentrations. Consequently, this study supports the need for reformulation of environmental legislation and the development of more protective public policies.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00128-025-04162-2.

Authors' Contributions Diego Ferreira Gomes: Conceptualization, Methodology, Formal analysis, Investigation, Writing; Thandy J. S. Pinto: Formal Analysis, Methodology and Review; Raquel Aparecida Moreira: Formal Analysis, Methodology, and Editing; Julia Gomes do Vale: Formal Analysis and Review; Hevelyn Plácido Gomes: Formal Analysis and Review; Michelly Pereira Soares: Formal Analysis and Review; Maíra Rodrigues Lima: Laboratory analysis; Marisa Narciso Fernandes: Laboratory analysis; Odete Rocha: Writing and Editing.

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Declarations

Competing interest The authors declare no competing financial interests or personal relationships that could have influenced this work.

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